

Artificial Intelligence in Drug Toxicity Prediction: Recent Advances, Challenges, and Future Perspectives

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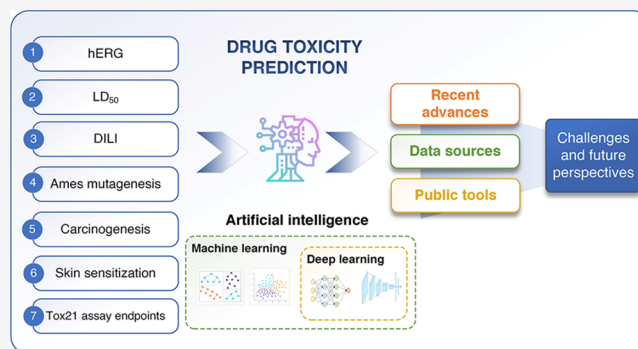
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ABSTRACT: Toxicity prediction is a critical step in the drug discovery process that helps identify and prioritize compounds with the greatest potential for safe and effective use in humans, while also reducing the risk of costly late-stage failures. It is estimated that over 30% of drug candidates are discarded owing to toxicity. Recently, artificial intelligence (AI) has been used to improve drug toxicity prediction as it provides more accurate and efficient methods for identifying the potentially toxic effects of new compounds before they are tested in human clinical trials, thus saving time and money. In this review, we present an overview of recent advances in AI-based drug toxicity prediction, including the use of various machine learning algorithms and deep learning architectures, of six major toxicity properties and Tox21 assay end points. Additionally, we provide a list of public data sources and useful toxicity prediction tools for the research community and highlight the challenges that must be addressed to enhance model performance. Finally, we discuss future perspectives for AI-based drug toxicity prediction. This review can aid researchers in understanding toxicity prediction and pave the way for new methods of drug discovery.

KEYWORDS: Artificial intelligence, Machine learning, Deep learning, Drug toxicity prediction, Drug discovery



1. INTRODUCTION

Toxicity is the extent to which a particular chemical substance or a mixture of substances can harm bodily systems or organs. Figure 1A illustrates examples of toxicity in several human organs. Predicting drug toxicity is an important step in the drug development process as it enables researchers to identify potential safety concerns and design drugs that are less likely to produce toxic effects. For a medicine to be authorized, it must meet certain safety and efficacy standards set by regulatory agencies such as the U.S. Food and Drug Administration (FDA) and European Medicines Agency (EMA).¹ It is estimated that over 30% of drug candidates are discarded to toxicity.^{1,2} Toxicity can also affect the marketability and commercial success of a drug.³

Traditionally, drug toxicity prediction has been performed by integrating in vitro and in vivo studies, wherein the effects of a drug on cells or animals are evaluated. This process of testing a drug for toxicity is labor-intensive, time-consuming, and involves extensive testing on animals and human cells. Until now, these methods have not been thought to be the most effective and humane.⁴ Toxicity predictions using rodent and nonrodent animal models only agree when extended to humans in 43% and 63% of cases, respectively, and in less than 30% of cases when predicting adverse drug reactions in target organs.⁵ Additionally, ethical concerns and the high cost of in

vivo animal testing have necessitated the development of viable alternatives. From this perspective, in silico toxicology methods offer an alternative option for assessing, modeling, and predicting the toxicity of new drugs and substances. Given their promising ability to reduce the use of animals for assessing toxicity end points, computational models present many benefits, including cost-effectiveness, rapid data availability, and standardization of procedures.

Artificial intelligence (AI) has become a trending topic in recent years for predicting drug toxicity in silico using computational models.⁶ Several studies have focused on using various machine learning (ML) and deep learning (DL) algorithms, such as neural networks, to analyze large data sets of drug and toxicity information to identify the potential toxic effects of drugs during development. These models have the potential to accelerate drug development significantly by identifying toxicities early in the development process, thereby avoiding the need for expensive and time-consuming animal

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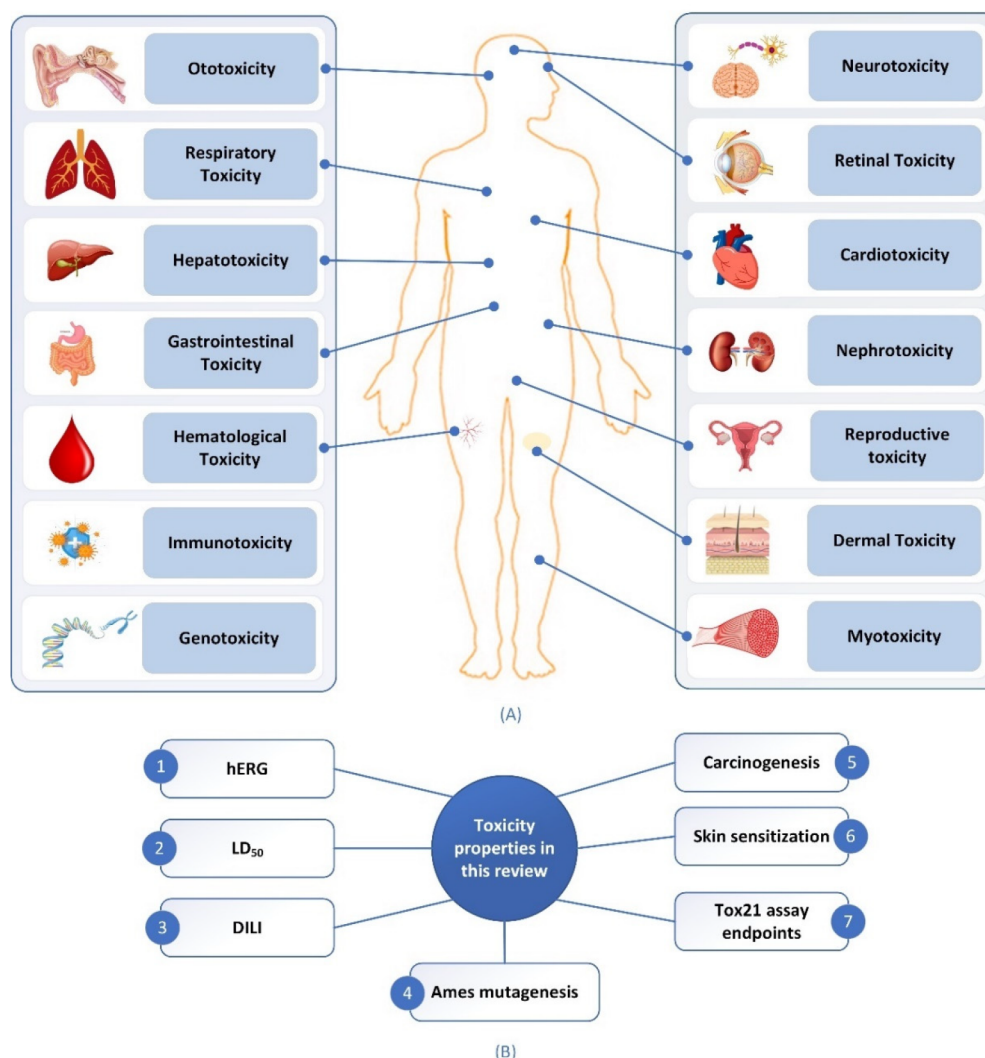


Figure 1. (A) Specific toxicity in some human organs and (B) toxicity properties in this review.

testing. Additionally, AI-based toxicity prediction models can be used to prioritize compounds for further testing and identify new drug targets and mechanisms of toxicity. Several reviews on the application of AI in toxicity prediction have been published;^{1,6–11} however, AI-based toxicity prediction is a broad field, involving diverse toxicity properties, and it is difficult to cover this area in its entirety in a single review. To develop, optimize, and enhance the performance of a model, it is essential to comprehensively explore the studies of toxicity properties and recent developments in AI-based toxicity prediction models. Here, we present a comprehensive review of recent advances in AI-based methods for drug toxicity prediction, including ML and DL, of six key toxicity properties and Tox21 assay endpoints (Figure 1B) from 2019 to the present. Moreover, we provide a list of public data sources and useful toxicity prediction tools for the research community and highlight the key challenges and future directions for AI-based drug toxicity studies.

2. RECENT ADVANCES IN AI-BASED DRUG TOXICITY PREDICTION

AI is a broad field that encompasses various techniques for teaching computers to perform tasks usually performed by

humans, such as pattern recognition, predictions, and decisions.^{12,13} ML is a type of AI that uses algorithms to learn from data, identify patterns, and make predictions. It can analyze large amounts of data and estimate the potentially toxic effects of drugs before they are evaluated in humans. Various ML methods, such as random forests (RF), decision trees (DT), k-nearest neighbor (k-NN), and support vector machines (SVM), have been used in drug toxicity prediction and have shown promising results in accurately identifying toxic compounds. DL is a specialized subfield of ML that uses neural networks to learn from data. Neural networks are complicated mathematical models that simulate functioning of the human brain. Neural networks in DL compose multiple layers that extract features from the input data and make predictions. Whereas traditional ML algorithms have fairly simple structures, such as linear regression (LR) or DT, DL is based on complex artificial neural networks (ANN), and thus requires larger amounts of data and more computing power. DL models have been used to predict drug toxicity and are highly effective for identifying toxic compounds and predicting the potentially toxic effects of drugs. Commonly used DL algorithms in recent years include deep neural networks (DNN), ANN, recurrent neural networks (RNN), convolutional neural networks (CNN), graph neural network (GNN),

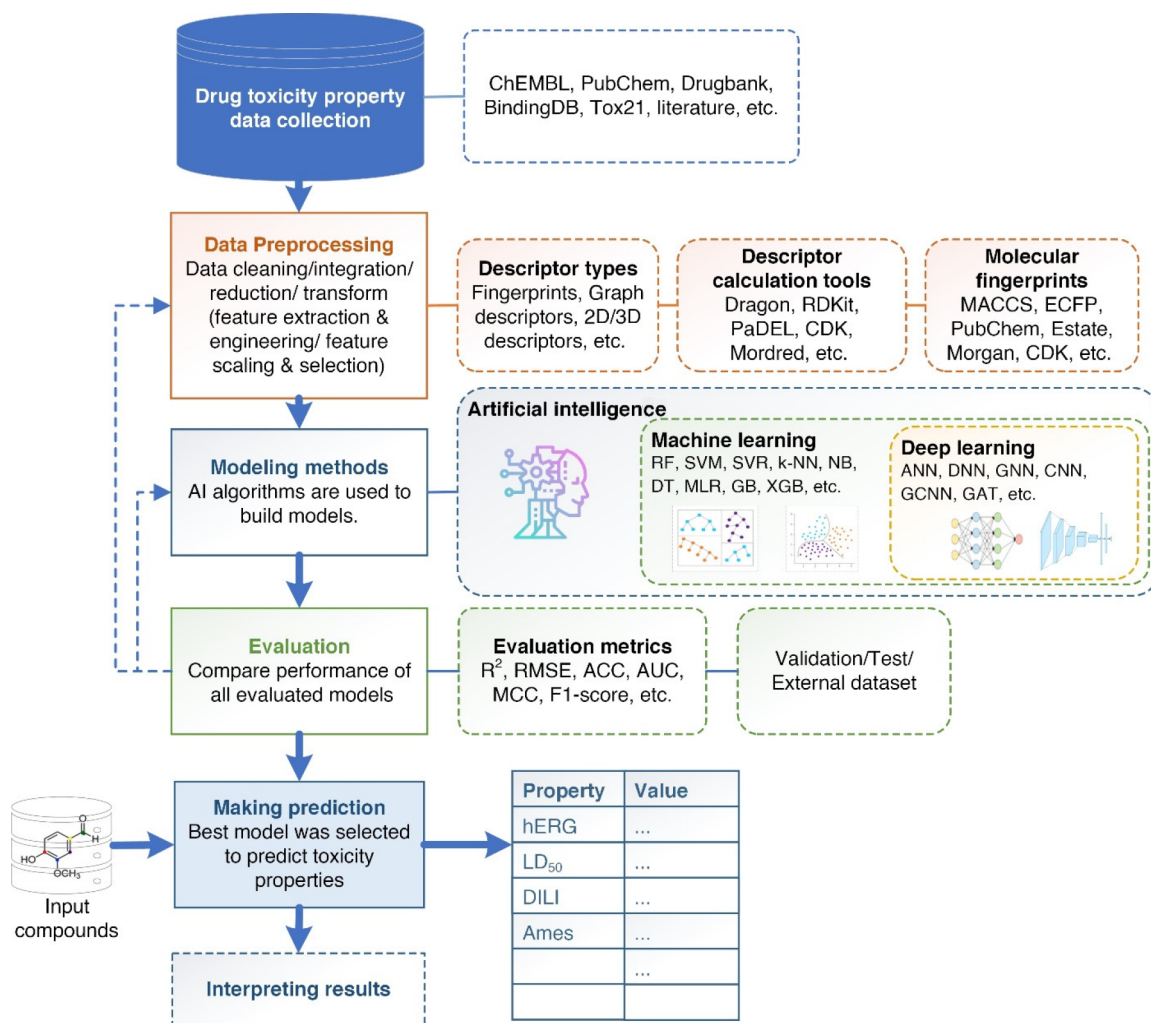


Figure 2. Process flow of drug toxicity prediction model.

graph convolutional neural networks (GCNN), and graph attention network (GAT).

In silico prediction of drug toxicity has received considerable research attention in recent years. The process flow of model prediction is shown in Figure 2. While there are numerous toxicity-related properties, in this review, we focused on some of the toxicity properties that have been investigated more frequently in recent years, such as human ether-a-go-go-related gene inhibition (hERG), lethal dose 50% (LD₅₀), drug-induced liver injury (DILI), Ames mutagenesis (Ames), skin sensitization, carcinogenesis, and Tox21 assay end points. We examine each toxicity property and explain why it was predicted, as well as the advancements in studies using AI that predicted it from 2019 to the present.

2.1. Cardiac Side Effect Prediction. The hERG gene (KCNH2) encodes the protein Kv11.1, which is the alpha subunit of the potassium ion channel. This ion channel is widely recognized for its role in the electrical activity of the heart. The hERG channel mediates polarized I_{Kr} currents in the action potentials of the heart, which in turn help to coordinate the heartbeat. Arrhythmia is one of the most prevalent adverse effects that contributes to drug rejection throughout pharmacological development.¹⁴ This failure is primarily related to the ability of the drug to inhibit the cardiac potassium channel hERG. Early identification of the hERG-

inhibitory properties of bioactive compounds has gained considerable attention over the years, and several in silico, in vitro, and in vivo trials have investigated the prevention of adverse cardiac events. The primary objective of these studies was to elucidate the processes underlying these effects and to provide suggestions or information to scientists involved in drug discovery and development that would help minimize cardiac side effects. The FDA and EMA highly recommend early studies on hERG toxicity to evaluate the predicted cardiovascular effects of drugs. This requires a large number of pharmacomodulation experiments on the newly discovered chemical series, including an evaluation of activity toward hERG, to be performed after a preliminary screening for hits. Table 1 summarizes several intriguing analyses conducted in recent years regarding various methods to relieve hERG toxicity, particularly by employing AI models.

Recently, ML-based approaches have increasingly been used in to predict hERG toxicity. Some ML models such as RF, SVM, naïve Bayes (NB), support vector regression (SVR), k-NN, DT, gradient boosting (GB), partial least-squares (PLS), and extreme gradient boosting (XGB) are widely used in hERG prediction. Venkatraman used extended connectivity fingerprint 6 (ECFP6)-based RF models to develop ADMET prediction.¹⁵ The accuracy (ACC) of the hERG prediction model was 80%, and the area under the receiver operating

Table 1. Summary of Recent Advances in hERG Prediction Using AI^a

Methods	Features	Data source	Data set size (compounds)	Performance	ref.
RF (BC)	FCFP6 FP	¹⁶	7,889	ACC = 0.80, ROC-AUC = 0.88	¹⁵
RF (BC)	2D descriptors	ChEMBL, PubChem	8,382	ACC = 0.932, MC C = 0.803	¹⁸
an extremely randomized tree algorithm (BC)	RDKit descriptors	ChEMBL	3,721	ACC = 0.801, Recall = 0.94	¹⁷
SVM (BC)	ECFP_4 FP, 2D, 3D descriptors	ChEMBL, GOSTAR, PubChem, and hERGCentral	291,000	ACC = 0.984, ROC-AUC = 0.962	¹⁹
RF, MLP& SMO (BC)	1D, 2D, 3D descriptors, 12 types of FP	ChEMBL v22, Pubchem, in-house, and ³¹	9,390	ACC = 0.93, S N = 0.967, S P = 0.78	²⁰
MT- DNN (BC)	MACCS FP	^{32–35}	7,889	ROC-AUC = 0.967, ACC = 0.925	¹⁶
DMPNN (BC)	9 types of FP	^{16,22}	14,409	ROC-AUC = 0.956	²¹
RF, NB, SVM, ANN, Bagging (BC)	FCFP_2, FCFP_4, FCFP_6 FP	ChEMBL	5,299	ACC = 0.91, ROC-AUC = 0.95	³⁶
RF, SVM, Dense neural network (BC)	2D descriptors	Drug Discovery Hackathon (the Govt. of India)	8,227	ACC = 0.96, ROC-AUC = 0.994	³⁷
RF, XGB, DNN, RNN (BC)	RDKit descriptors	ChEMBL v.25 and NCATS	~9,000	ACC = 0.80, ROC-AUC = 0.84	³⁸
DNN, RF, XGB, Consensus model (BC)	MACCS and Morgan FP	ChEMBL and ¹⁶	12,850	ROC-AUC = 0.908, ACC = 0.84	²³
Restricted Boltzmann Machine-Capsule Network and Convolution-Capsule Network (BC)	MACCS FP, 13 descriptors	³²	2,644	ACC = 0.918, ROC-AUC = 0.956	²⁴
MGAF (BC)	Graph structure	ChEMBL, PubChem, OCHEM, literature	13,845	ACC = 0.889, ROC-AUC = 0.943	³⁰
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	8,672	ACC = 0.919, ROC-AUC = 0.784	²⁹
Fully connected neural network, GCNN, CNN (BC)	Morgan FP, 2D and 3D physicochemical descriptors, Graph structure	BindingDB, ChEMBL, and ^{16,32,34}	12,620	ACC = 0.86, ROC-AUC = 0.93	²²
DNN, GCNN (BC)	Mordred descriptors, ECFP2, PubChem FP, Graph structure	BindingDB, ³⁹ ChEMBL, ⁴⁰ in-house, and ^{16,32,34,41}	14,440	ACC = 0.829, ROC-AUC = 0.896	²⁵

^aBC, binary classification (hERG blocker/nonblocker); FP, fingerprint.

characteristic curve (ROC-AUC) was 88% for 7,889 compounds. Using the same data set, deepHERG models yielded a higher prediction ACC of 92% and a ROC-AUC of 96.7% when using a multitask (MT)-DNN.¹⁶ Additionally, the RF model was used in other hERG prediction studies by Hsiao et al.¹⁷ and Arab et al.,¹⁸ and achieved remarkable ACC prediction results. Ogura et al. built an SVM model for hERG inhibitor prediction using an integrated large hERG data set containing over 291,000 compounds from ChEMBL, PubChem, GOSTAR, and hERGCentral databases.¹⁹ This model used ECFP_4 structural fingerprints and 72 descriptors selected by NSGA-II and achieved an ACC of 98.4% and a kappa statistic of 0.733. The performance of this model was superior to that of commercial models (ADMET Predictor, ACD/Percepta, and StarDrop) on the test set. Konda et al. developed hERG classification models using RF, sequential minimal optimization (SMO), and multilayer perceptron (MLP) algorithms and a curated data set of 8,705 compounds from ChEMBL v.22 with two-dimensional (2D) descriptors.²⁰ Their consensus model performance improved from 16% to 52% in predicting hERG inhibitors compared to other models, such as PaDEL-DDPredictor v2.1, Simulation plus/ADMET Predictor v8.1, Schrödinger/QikProp, StarDrop v6.4, admetSAR, CoFFer, pkCSM, and pred-hERG with an ACC of 92%.

Furthermore, the DL model has been widely used in building hERG predictive models, such as DNN, ANN, RNN, CNN, GNN, GCNN, and GAT. Recently, Shan et al. used a directed message-passing neural network (DMPNN) model to predict hERG blockers.²¹ They evaluated various combinations

of the DMPNN model with many descriptors and fingerprints, such as MACCS, PubchemFP, ECFP4, ECFP6, FCFP4, Rdkit 2D normalized, Mol2vec, MOE53, and moe206. The DMPNN combined with moe206 descriptors demonstrated the best performance for 7,889 compounds, with a ROC-AUC value of 0.956 in 5-fold cross-validation. However, it performed slightly worse than CardioTox on Karim's test sets,²² with an ACC of 80% (compared with the 81% ACC of CardioTox). Zhang et al. developed an hERG blocker classification model called HergSPred using a consensus model obtained by calculating the mean of the three prediction methods, namely DNN, RF, and XGB models based on the Morgan and MACCS fingerprints.²³ HergSPred outperformed earlier models, notably those of Cai et al.¹⁶ and Wang et al.²⁴ when identical training and test sets were used. Specifically, with the same training data set of 3,485 compounds and test data set of 435 compounds, HergSPred achieved an ACC of 99.2%, compared with the 92.5% ACC of the model of Cai et al. Furthermore, using the same training data set of 1,112 compounds and a test data set of 108 compounds, HergSPred achieved an ACC of 98.3% compared with the 91.8% ACC of the model of Wang et al.

DeepHit is a DL framework built by Ryu et al. to predict hERG using graph-based GCNN, descriptor-based DNN, and fingerprint-based DNN models.²⁵ DeepHit was evaluated using a large data set of 7,808 hERG nonblockers and 6,632 hERG blockers. On an external test data set, DeepHIT outperformed currently available tools, such as CardPred,²⁶ OCHEM Predictor,²⁷ and Pred-hERG 4.2²⁸ in terms of ACC (0.773),

Table 2. Summary of Recent Advances in LD₅₀ Prediction Using AI^a

Methods	Features	Data sources	Data set size (compounds)	Performance	ref.
RF (MC)	PUBCHEM FP	OCHEM ⁴⁸	11,363	ACC = 0.68, ROC-AUC = 0.81	15
k-NN, NB, Binned-Nearest Neighbors, Consensus (BC)	ECFPs FP	48,58	11,992	SE = 0.81, SP = 0.84	47
Integrated model: RF in regression, balanced RF, aiQSAR, istkNN, HPT-FR (BC, MC, and regression)	Dragon descriptors	NICEATM and NCCT	8,448	R ² = 0.716, RMSE = 0.477	48
aiQSAR method (MC)	Dragon descriptors	NICEATM and NCCT	8,890	ACC = 0.63	49
Multifingerprints similarity approach (BC, MC)	19 types of FP	58	8,944	MCC = 0.924, ACC = 0.964	59
MT-DNN, Single-task-DNN, RF, DL consensus architecture, GCNN (regression)	Morgan FP, RDKit descriptors	ChemIDplus ⁶⁰	80,081	R ² = 0.57, RMSE = 0.65	50
Algebraic graph-assisted bidirectional transformers model (regression)	Graph structure	55	7,413	RMSE = 0.570	61
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	7,334	R ² = 0.575, RMSE = 0.106	29
MolGIN method (regression)	Graph structure	53	7,413	R ² = 0.622, RMSE = 0.575	51
BTAMD model (regression)	Daylight, MACCS, Estate 1, Estate 2 FP	53	7,413	R ² = 0.662, RMSE = 0.557	52
CNN (regression)	Binary-Encoded SMILES (BES)	55	7,413	R ² = 0.619, RMSE = 0.603	57
DT and shallow neural network (regression)	2D descriptors	55	7,413	R ² = 0.629	56
FFNN, GCNN, CNN (regression)	2D, 3D, graph descriptors, ECFPs	55	7,380	R ² = 0.687, RMSE = 0.537	54

^aBC, binary classification; MC, multiclass classification; FP, fingerprint.

Matthew's correlation coefficient (MCC) (0.476), and sensitivity (SE) (0.833). In 2022, Wei et al. developed an ADMET prediction web server using GCNN and GAT called Interpretable-ADMET.²⁹ The hERG prediction model of Interpretable-ADMET was evaluated on 8,672 compounds and achieved the highest ACC (91.9%) and ROC-AUC (78.4%). ADMETLab 2.0³⁰ using a multitask graph attention framework (MGAF) to predict ADMET properties yielded the best ACC and ROC-AUC of 88.9% and 94.3%, respectively, on a large hERG data set of 13,845 compounds.

2.2. LD₅₀ Prediction. In toxicology, LD₅₀ is a standardized measure for identifying and comparing the toxicity of chemicals. The LD₅₀ value is the amount of a chemical needed to eliminate half (50%) a set of test animals tested after a particular testing period. In drug screening, the determination of LD₅₀ is often the first step in estimating and evaluating the toxic properties of a substance.⁴² Previously, the rodent acute oral toxicity test was predicated on the calculation of LD₅₀. However, owing to the limitations arising from interspecies variability and various ethical concerns, new testing methodologies have supplanted classic LD₅₀ testing.⁴³ Alternative methods are being explored to replace acute toxicity assays in animals, including in vitro (tissue culture) and in silico assessment and prediction of LD₅₀.⁴⁴ Table 2 summarizes some methods of using AI in LD₅₀ prediction research conducted since 2019. The binary classification model classifies chemicals into two categories: toxic (e.g., LD₅₀ ≤ 50 mg/kg) and nontoxic (e.g., LD₅₀ > 2000 mg/kg). The multiclass categorization of compounds is generally based on the U.S. Environmental Protection Agency⁴⁵ (LD₅₀ ≤ 50, 50 < LD₅₀ ≤ 500, 500 < LD₅₀ ≤ 5000, LD₅₀ > 5000 mg/kg) and Globally Harmonized System of Classification and Labeling of Chemicals schemes⁴⁶ (LD₅₀ < 5, 5 ≤ LD₅₀ < 50, 50 ≤ LD₅₀ < 300, 300 ≤ LD₅₀ < 2000, LD₅₀ ≥ 2000).

Recently, two new software tools for ADMET prediction have become available, namely, FP-ADMET¹⁵ and Interpretable-ADMET,²⁹ both of which have built-in LD₅₀ prediction

models. The LD₅₀ prediction models were developed Ballabio et al. using NB, N-Nearest Neighbors, Binned-Nearest Neighbors algorithms, and Extended Connectivity binary fingerprints.⁴⁷ The models used a large data set of 8,992 chemicals and achieved the highest specificity and sensitivity values of 84% and 81%, respectively, for the test set. Gadaleta et al. developed a QSAR model to predict a large data set containing 8,448 compounds.⁴⁸ They used balanced RF, RF in regression, the aiQSAR model,⁴⁹ the istkNN tool, the SARpy model, RF with hyperparameter tuning, and a generalized linear model to build an integrated model. The best integrated model achieved a root mean squared error (RMSE) of 0.477, and the best classification model achieved a balanced ACC greater than 70% for multiclass and greater than 80% for binary end points.

Jain et al.⁵⁰ constructed a consensus of multitasking DL methods utilizing RF, DNN, CNN, and GCNN, and a large data set of 80,081 compounds. The best performance of their consensus model achieved an RMSE of 0.65 and an R² of 0.57 over 5-fold cross-validation, which was better than that of the others in their study. The MolGIN model by Peng et al.⁵¹ and the boosting tree-assisted multitask DL (BTAMD) model by Jiang et al.⁵² predicted LD₅₀ using the same data set of 7,413 compounds from Zhu's data set.⁵³ The BTAMD model had a slightly higher predictive ACC (RMSE = 0.557, R² = 0.662) than the MolGIN model (RMSE = 0.575, R² = 0.622). Karim et al. proposed a DL framework for quantitative toxicity prediction, called QuantitativeTox, using a feed-forward neural network (FFNN), CNN, GCNN, and their own baseline feature representations.⁵⁴ Their models were evaluated using the TopTox data set⁵⁵ from the ECOTOX aquatic toxicity database, which contained 7,413 compounds for LD₅₀ prediction. QuantitativeTox achieved a higher performance than TopTox, a single-task hybrid shallow neural network, and the DT models by Karim et al.,⁵⁶ and BESTox,⁵⁷ with R² values of 0.687, 0.653, 0.629, and 0.619, respectively.

Table 3. Summary of Recent Advances in DILI Prediction Using AI^a

Methods	Features	Data sources	Data set size (compounds)	Performance	ref.
RF (BC)	PUBCHEM FP	⁷¹	2,478	ACC = 0.79, ROC-AUC = 0.88	¹⁵
LoR, RF (BC)	5 types of descriptors, ECFP6, PubChem, standard FP	DrugDex, DrugPoints, and DailyMed ⁸⁵	902	ROC-AUC = 0.8	⁶⁹
LoR, SVM, GB, AB, XGB, RF, ET, LightGBM, Catboost (BC)	12 types of FP and 7 descriptors	⁸⁶	450	ACC = 0.817, SE = 0.646 SP = 0.962	⁷⁰
Ensemble Models: k-NN, MLP, RF, NB, SVM, LoR, classification tree, FLDA, Bayes network (BC)	QuBiLS-MAS 0–2.5D molecular descriptors	⁷⁵	1,195	ACC = 0.84, SE = 0.89 SP = 0.761	⁷¹
NB, k-NN, RF, AB, Bagging, KStar, J48, DI4j, Vote (BC)	55 descriptors	^{75,86–98}	1,416	ACC = 0.783, ROC-AUC = 0.859	⁷²
Consensus model: SVM, HQPSO (BC)	PubChem, subFP, KRFP, Estate FP, 1D, 2D PaDEL descriptors	^{90,91,99–101}	2,608	ACC = 0.784, ROC-AUC = 0.829	⁷³
Bayesian, RF, k-NN, SVM, NB, AB, DL (BC)	ECFP6 FP	DILIRank, ⁸⁶ BDDCS, ¹⁰² Pfizer, AstraZeneca	2,191	ROC-AUC = 0.814, ACC = 0.746	⁷⁴
MGAF (BC)	Graph structure	ChEMBL, PubChem, OCHEM, literature	467	ACC = 0.894, ROC-AUC = 0.924	³⁰
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	644	ACC = 0.723, ROC-AUC = 0.718	²⁹
DeepDILI Model: NN, LR, k-NN, SVM, RF, XGB (BC)	Mold2 descriptors	¹⁰³	1,002	ACC = 0.687, ROC-AUC = 0.659	⁷⁶
GNN (BC)	Graph structure	^{86,104–107}	15,675	ACC = 0.814, ROC-AUC = 0.882	⁷⁷
GNN (BC)	Graph structure	CMap and LTKB	12,348	ACC = 0.773, MCC = 0.5525	⁷⁸
ANN, k-NN, LR, RFR, SVM (BC)	Physicochemical descriptors, ChemoTyper FP	⁸⁶	1,036	ACC = 0.82, MCC = 0.639	⁷⁹
ResNet18 (BC)	image features	^{86,94,97,100,108–110}	1,446	ACC = 0.958, F1 score = 0.926	⁸⁰
CNN (BC)	Morgan FP	^{75,86,101,111,112}	1,919	ACC = 0.89, ROC-AUC = 0.96	⁸¹
DL, SVM (BC)	1574 feature genes	^{113,114}	988	ACC = 0.971, ROC-AUC = 0.989	⁸³

^aBC, binary classification (positive DILI and negative DILI); FP, fingerprint.

2.3. DILI Prediction. DILI is an unfavorable reaction to medicines or other xenobiotics that occurs either predictably or unexpectedly when an organism is exposed to toxic doses of specific chemicals.⁶² DILI has long been a focus of interest for doctors and drug researchers. Determining the effectiveness of a DILI prediction model in terms of reliably predicting liver injury is a pressing issue for drug researchers. DILI is the most frequently mentioned reason for market withdrawal (up to 32% of drug recalls);⁶³ however, accurate prediction of human DILI using cytotoxicity screening in vitro or in animal studies is difficult and complex. AI approaches, especially those based on ML, range from conventional ML approaches such as RF, k-NN, and SVM to more recent techniques like DL, such as DNN, CNN, and GNN, and provide tools that can analyze the chemistry of compounds and accurately predict some of their properties based on their structure alone. Many notable reviews have been conducted in recent years regarding different approaches to predict DILI in silico.^{64–67}

From 2019 to 2022, many studies have used ML methods such as RF, LR, NB, SVM, k-NN, AB, GDBT, and ET, to develop a DILI prediction model with high predictive efficiency^{15,68–74} (Table 3). The most notable is the ensemble model based on QuBiLS-MAS Features and Shallow Learning by Mora et al. using k-NN, MLP, RF, NB, SVM, logistic regression (LoR), classification tree, Fisher's linear discriminant analysis (FLDA), and Bayes network algorithms.⁷¹ The best performance of their model achieved an ACC of 84% over 10-fold cross-validation using a training set of 1,075

compounds from a previous study.⁷⁵ Wang and Chen constructed five consensus models with PaDEL descriptors and substructures, Klekota–Roth, Estate, and PubChem fingerprints using SVM and hybrid quantum particle swarm optimization (HQPSO) algorithms and a data set of 2,608 chemicals.⁷³ Their best consensus model correctly predicted 99.2% of the 528 test chemicals, with an ACC of 80%, SE of 83.9%, and specificity (SP) of 73.3%. Recently, three new software tools for ADMET prediction have become available, namely, ADMETLab 2.0,³⁰ FP-ADMET,¹⁵ and Interpretable-ADMET,²⁹ all of which have a built-in DILI prediction model with a DILI prediction ACC > 79%.

Furthermore, many studies have used the DL method to improve the accuracy of DILI prediction.^{29,30,76–83} Li et al. designed a DeepDILI model using a DNN model based on Mold2 descriptors⁷⁶ (Table 3). The DeepDILI model outperformed five ML algorithms (RF, k-NN, SVM, LR, and XGB) and two advanced ensemble methods, with an ACC of 68.7% and an MCC value of 0.331 on a data set of 1,002 drugs. Hwang et al. developed a DILI prediction model using Transcriptional Response Data with GNN called GLIT,⁷⁸ which was evaluated on 12,348 samples from the LTKB and CMap data sets. GLIT outperformed the baseline models, including MLP, GCNN, and GraphSAGE,⁸⁴ with an ACC of 77.3% and MCC of 0.553. In another study, Ma et al. used a multiview GNN with attribute increments from four toxicity data sets, increasing the number of compounds to 15,675 for DILI prediction, improving accuracy from 78.8% to 81.4% and

Table 4. Summary of Recent Advances in Ames Mutagenesis Prediction Using AI^a

Methods	Features	Data sources	Data set size (compounds)	Performance	ref.
RF (BC)	PUBCHEM FP	¹²²	7,950	ACC = 0.79, ROC-AUC = 0.87	¹⁵
aiQSAR method: RF, ET, GLMBoost, SVM (BC)	Dragon descriptors	¹²³	5,764	ACC = 0.81	⁴⁹
PLSDA, MDA, SVM, k-NN, RF, XGB, GBM, C5 (BC)	MP, ECFP, FCFP FP	¹²²	7,617	ROC-AUC = 0.84, ACC = 0.79	¹¹⁶
RF, GBT, k-NN (BC)	Morgan FP	CCRIS, GENE-TOX, NTP	5,061	F1 score = 0.82, MCC = 0.65	⁶⁸
DNN, SVM, k-NN, RF (BC)	alvaDesc descriptors	^{122,124–127}	4,053	ACC = 0.838, ROC-AUC = 0.894	¹¹⁸
LightGBM, RF, SVM, XGB, DNN (BC)	Morgan FP, RDKit, MolVS descriptors	¹⁰⁷	6,509	ROC-AUC = 0.811, ACC = 0.811	¹²⁸
MolGIN method (BC)	Graph structure	¹²²	7,619	ROC-AUC = 0.918, ACC = 0.839	⁵¹
GNN (BC)	2D graph structure	¹¹⁵	7,617	ROC-AUC = 0.92	¹¹⁹
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	7,387	ACC = 0.843, ROC-AUC = 0.842	²⁹
MGAF (BC)	Graph structure	ChEMBL, PubChem, OCHEM, literature	7,575	ACC = 0.807, ROC-AUC = 0.902	³⁰
GCNN (BC)	2D Graph structure	NIHS ¹⁰⁷	19,494	ACC = 0.85, ROC-AUC = 0.88	¹²⁰
GCNN (BC)	RDKit descriptors	¹⁰⁷	6,307	ROC-AUC = 0.878, ACC = 0.81	¹²¹

^aBC: binary classification (mutagen and nonmutagen); FP: fingerprint.

ROC-AUC from 86.6% to 88.2%.⁷⁷ Nguyen-Vo et al. built a DILI prediction model using CNN algorithms and molecular fingerprint-embedded features on a development set of 1,597 compounds and a test set of 322 compounds.⁸¹ Their study yielded a high ACC of 89% and ROC-AUC of 96%. An approach published in 2022 called ResNet18DNN also used a CNN model that achieved a very high prediction ACC of 95.8% on a data set of 1,446 compounds.⁸⁰

2.4. Ames Mutagenesis Prediction. The Ames test is a widely used biological test wherein bacteria are used to evaluate the mutagenicity of chemical compounds.¹¹⁵ The mutagenic effect of chemicals is closely related to carcinogenicity, and the Ames test is the most widely used method for testing the mutagenicity of compounds.³⁰ The identification of mutagenic compounds is a critical task in the early stages of drug discovery. Nevertheless, the experimental procedure of mutation identification using the Ames test is expensive and time-consuming because the compounds must be synthesized and evaluated, and the results are often neither precise nor reproducible.¹¹⁶ Therefore, constructing strong AI-based models that can accurately predict the mutagenicity of a compound before synthesis can help address the limitations of the Ames test.

Multiple methods of constructing mutagenicity prediction models have been developed over the last 20 years, and both ML and DL methods have shown desirable performance and reliability. The most recent in silico mutagenicity prediction studies have utilized the strengths of large data sets and achieved high ACC ranging from 79% to 85% (Table 4). Many ML and DL algorithms, such as RF, SVM, k-NN, XGB, GBT, GNN, and GCNN, have been used to build predictive models of mutagenicity instead of in vitro tests.^{15,29,30,51,68,116–121} Chu et al.¹¹⁶ used eight ML algorithms, including RF, SVM, XGB, partial least-squares discriminant analysis (PLSDA), mixture discriminant analysis (MDA), SVM, k-NN, and C5, to predict the mutagenicity of 7,617 compounds from a previous study.¹²² By combining a variety of molecular fingerprints

and physicochemical molecular properties as compound descriptors with a selection of alternative modeling algorithms, models with good predictive ability were found that offered molecular insights and revealed aspects of molecules that cause mutagenicity. The RF model exhibited the highest performance with the test set, with a balanced ACC of 79% and a ROC-AUC of 84%. The RF model of FP-ADMET had a similarly accurate Ames prediction when using the same data set.¹⁵ The enhanced graph isomorphism network model developed by Peng et al. achieved superior predictive performance with an ACC of 83.9% and a ROC-AUC of 91.8%⁵¹ when using the data set of Xu et al.

Vukovic et al. proposed a novel method called aiQSAR based on fingerprints to predict Ames on a data set of 5,764 compounds, achieving an ACC of 81%.⁴⁹ For comparison with traditional ML, Kumar et al. built mutagenicity prediction models using DNN, SVM, RF, and k-NN.¹¹⁸ Their models were evaluated on 5,053 compounds, with 1,597 molecular feature vectors encoded for each compound. The DNN-based prediction model outperformed traditional methods such as RF, SVM, and k-NN, with a prediction ACC of 83.1% and a ROC-AUC of 83.8% for the test set. Recently, Guo et al. used the GNN framework to build a multilayer self-attention method called Ligandformer for compound property prediction, such as aqueous solubility, Caco-2 cell permeability, and Ames mutagenesis.¹¹⁹ On a data set of 7,617 compounds, their Ames prediction model made highly accurate predictions with a ROC-AUC of 92%. Hung and Gini developed a mutagenicity prediction model using a GCNN based on 2D molecular graphs without descriptors.¹²⁰ Their best model achieved a predictive ACC of 85% and an ROC-AUC of 88% using a large data set containing over 19,000 compounds. Several recent studies^{29,30,121} also used the GCNN on large data sets, and they all provided remarkably high ACC prediction results.

2.5. Carcinogenesis Prediction. Carcinogenicity is one of the most highly scrutinized toxicological end points of

Table 5. Summary of Recent Advances in Carcinogenesis Prediction Using AI^a

Methods	Features	Data sources	Data set size (compounds)	Performance	ref.
RF (BC)	PUBCHEM FP	¹³⁵	1,003	ACC = 0.68, ROC-AUC = 0.75	¹⁵
Hybrid: CNN, MLP, FFNN (BC)	2D molecular descriptors	CCRIS, DSSTox, NIOSH, JSOH, IARC, and NTP ^{135,137}	7,994	ACC = 0.743, ROC-AUC = 0.806	¹³⁰
			1,618	ACC = 0.703, ROC-AUC = 0.725	
CapsCarcino model (BC)	6MD, 13MD descriptors, 7 types of FP	¹³⁵	1,003	MCC = 0.378, ACC = 0.745	¹³¹
DeepCarc Model: DNN (BC)	Mol2vec, Mold2, MACCS descriptors	¹³⁸	863	ACC = 0.754, ROC-AUC = 0.776	¹³²
GNN (BC)	Graph structure	CPDB, ¹³⁹ CCRIS ¹⁴⁰	5,500	ROC-AUC = 0.73	¹³³
MGAF (BC)	Graph structure	ChEMBL, PubChem, OCHEM, literature	1,041	ACC = 0.619, ROC-AUC = 0.694	³⁰
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	1,146	ACC = 0.716, ROC-AUC = 0.69	²⁹
Cropping GCNN (BC)	Graph structure	Predictive Toxicology Challenge ¹⁴¹	1,380	ACC = 0.694	¹³⁴

^aBC: binary classification; FP: fingerprint.Table 6. Summary of Recent Advances in Skin Sensitization Prediction Using AI^a

Methods	Features	Data sources	Data set size (compounds)	Performance	ref.
RF (BC)	MACCS FP	¹⁴²	1,278	ACC = 0.74, MCC = 0.49	¹⁴⁵
RF (BC)	MACCS FP	^{148,149}	1,285	ACC = 0.75, MCC = 0.51	¹⁴²
RF (BC)	PUBCHEM FP	¹⁴⁶	1,033	ACC = 0.67, ROC-AUC = 0.74	¹⁵
RF and SVM (BC)	Morgan2 FP	^{149,150}	1,416	ACC = 0.81, ROC-AUC = 0.89	¹⁴⁷
MGAF (BC)	Graph structure	ChEMBL, PubChem, OCHEM, literature	405	ACC = 0.854, ROC-AUC = 0.901	³⁰
GCNN, GAT (BC)	Graph structure	ChEMBL, PubChem, DrugBank, literature	512	ACC = 0.894, ROC-AUC = 0.872	²⁹
NB (BC)	ECFP4, Morgan2 FB	Human, LLNA, KeratinoSens, h-CLAT, DPRA	1,682	SE = 0.94, SP = 0.84	¹⁴⁶

^aBC, binary classification (carcinogens and noncarcinogens); FP, fingerprint.

chemical compounds because of its detrimental impact on human health.³⁰ According to some theories, the ability of chemicals to destroy DNA or disrupt cellular metabolic processes may lead to cancer. Many FDA-approved medications have been withdrawn from the market after being identified as carcinogenic to humans and animals. Screening for potentially carcinogenic compounds is crucial for preventing environmental cancers.¹²⁹ In this domain, a large number of prediction models have been evaluated, ranging from short-term biological studies (e.g., mutagenicity tests) to theoretical models. In silico theoretical techniques, such as ML and DL, are highly desirable for carcinogen detection because they actively support the substitution, reduction, and enhancement of animal studies.

In recent years, many AI-based models and tools have been developed to predict the carcinogenic potential of compounds.^{15,29,30,130–134} Limbu and Dakshanamurthy built carcinogenicity prediction models using a hybrid NN architecture with three data sets containing more than 10,000 chemicals, and 653 molecular descriptors.¹³⁰ The hybrid NN achieved an average ACC of 74.3% and mean ROC-AUC of 80.6%, which were superior to those of the AdaBoost model but not those of the Bagging and RF models using the same data. One of the leading causes of low predictive accuracy of DL models is sparse data sets. To solve this problem, a new DL architecture, CapsCarcino, was developed by Wang et al. to predict compound carcinogenicity.¹³¹ Their model, which was evaluated on a training data

set of 1,003 compounds from a previous study¹³⁵ using 5-fold cross-validation, achieved better prediction performance than other methods (SVM, RF, k-NN, XGB, and CNN), with an average ACC of 74.5%, and achieved an ACC prediction rate of 83% on an external validation data set. Furthermore, CapsCarcino performed well with other models when evaluated on a sparse training data set randomly reduced to 20%, 40%, 60%, and 80% of the original full training data set. The DeepCarc model is also a good DL model for predicting carcinogenicity and was recently developed by Li et al.,¹³² using a data set of 863 compounds and three descriptors: Mol2vec, Mold2, and MACCS. The DeepCarc model performed better than other models, such as RF, k-NN, SVM, and XGB, using the test set, with an ACC of 75.4% and an average improvement rate of 37%. Details of other models are listed in Table 5 and other helpful reviews.^{129,136}

2.6. Skin Sensitization Prediction. The skin sensitization potency is an important end point for assessing the safety of novel compounds and chemical combinations.¹⁴² Commercial items, such as ointments or cosmetics, must comply with local safety standards for animal and nonanimal models, including allergies. However, in Europe, animal testing for determining the safety of cosmetic components has been prohibited since 2013, and other nations have followed suit.¹⁴³ Consequently, a series of nonanimal in vitro assays for the determination of skin sensitization, such as the direct peptide response assay, KeratinoSens, LuSens, and the h-CLAT and U-SENS human cell line activation assays, were developed and approved in

OECD test guidelines. Furthermore, multiple AI-based models for predicting skin sensitization have been constructed using diverse modeling approaches based on animal and nonanimal data sets. The skin sensitization hazard assessment framework has been outlined in a previous review that covers both experimental and *in silico* results.¹⁴⁴

In recent years, several AI-based models for predicting skin sensitivity to small organic molecules have been published^{15,29,30,142,145–147} (Table 6). Wilm et al. developed a skin sensitization prediction model called Skin Doctor CP based on RF algorithms using a data set of 1,285 compounds from two previous studies^{148,149} with MACCS descriptors.¹⁴² The Skin Doctor CP model had an ACC of 75%, MCC of 0.51, and efficiency of 92% for a test set at a level of 0.3. These results show that the Skin Doctor CP model is better than the previous Skin Doctor CP: Bio.¹⁴⁵ In addition, the Skin Doctor model previously built by this group, which used a combination of RF and SVM classifiers with multiple fingerprints and molecular descriptors, achieved a highly accurate prediction result of 81% and an ROC-AUC of 89%.¹⁴⁷ Pred-Skin v.3.0 is a free, fast, and user-friendly web portal for predicting human skin sensitization.¹⁴⁶ Furthermore, several other tools for predicting ADMET, such as FP-ADMET,¹⁵ ADMETLab 2.0,³⁰ and Interpretable-ADMET,²⁹ also integrate accurate skin sensitivity predictions.

2.7. Tox21 Assay End Point Prediction. The Toxicology in the Twenty-First Century (Tox21) program, a government collaboration between the National Institutes of Health, the Environmental Protection Agency, and the FDA, aims to improve toxicity evaluation methods.^{151,152} The objective of the program was to rapidly and efficiently determine whether certain chemical compounds have the potential to alter human physiological systems, resulting in detrimental health effects. In the Tox21 Challenge, participants are allowed to join 15 different subchallenges. The Tox21 data set contains 12,060 training and 647 test samples, representing chemical molecules with 12 different toxic effects. Although this challenge was launched in 2008, it continues to attract participants. The Tox21 data set is useful for *in silico* toxicology studies.

In recent years, significant progress has been made in developing powerful AI models for toxicity prediction using the Tox21 data set. Supporting Information Table S1 summarizes AI-based studies on predicting toxicity end points using the Tox21 data set, focusing on those published from 2019 to the present. While some approaches used ML methods, such as RF, GBDT, and SVM,^{153,154} the majority used DL methods, such as NN, DNN, RNN, CNN, GNN, and GCNN.^{29,30,51,128,155–164} Most models achieved highly accurate predictions, with an ACC of up to 99.5% and ROC-AUC of up to 99.1%.

We have summarized the *in silico* models published from 2019 to the present for predicting the necessary and common toxicity properties. We believe that these advances using ML and DL models to predict toxicity *in silico* will make molecular screening more reliable, which in turn will reduce the time, cost, and requirement of animal testing to some extent. These next-generation models not only enable improved diagnosis of drug effects earlier in the identification process but will also assist regulators in prioritizing compounds for testing with careful consideration of potential harmful effects.

3. DATA SOURCES FOR AI-BASED DRUG TOXICITY PREDICTION RESEARCH

When using AI for drug toxicity prediction, the choice of database is particularly important because the quality and diversity of the data can have a substantial impact on the performance of the predictive models. Several additional factors should be considered when selecting a database for AI-based drug toxicity predictions.

Data Diversity. As with any predictive modeling, it is important to have a diverse set of data to train the model to enable generalization to new compounds.¹⁶⁵ This is particularly important for drug toxicity prediction, in which the goal is to determine the toxicity of new compounds that have not been previously reported. Whether the data includes a wide range of chemical structures, mechanisms of toxicity, and end points are important factors to be considered.

Data Bias. Data bias refers to the systematic errors in a data set that can lead to incorrect or unfair conclusions. There are several potential sources of data bias in toxicity predictions, including imbalanced data, missing data, and sampling bias. Over time, many biases that can artificially increase or decrease prediction performance have been identified.¹⁶⁶ For example, data imbalances may have caused performance overestimation with an ROC-AUC of 0.962 in the study of Ogura et al., who used a large data set containing 9,890 inhibitors and 281,329 inactive compounds to train their models.¹⁹ In another study,³⁷ the authors used an imbalanced data set (6,878 blocker compounds/1,349 nonblocker compounds) that can lead to high bias and performance overestimation, with an ROC-AUC of 0.994. Although they solved this problem using the data augmentation method, it is difficult to guarantee that the augmented data are reliable because in contrast to an augmented data picture, they cannot be easily checked for validity. When working with imbalanced data sets, it is recommended that authors report the MCC or the area under the precision-recall (PR) curve rather than the ROC-AUC.¹⁶⁷ The PR curve can be plotted and should be useful to visualize the screening power of a method. However, when conducting virtual screening, it is crucial to prioritize the early enrichment of true positives among the top-ranked instances. To accurately assess the early enrichment, authors should report the enrichment factor of true positives in the top 1% of instances (EF1%) and its normalized version (NEF1%) as indicators.

Data Balance.¹⁶⁸ It is important to consider whether the data are balanced in terms of the number of positive and negative examples of toxicity, which will help the model learn the underlying patterns of toxicity and thus avoid over- or under-generalization.

Data Annotation. It is important to consider whether the data are annotated and labeled correctly, and whether they can be used in a supervised learning setup. To enhance the predictive capabilities of the model, we also considered whether the data were multimodal, such as images, transcriptomics, and metabolomics.

Data Scalability. It is important to consider whether the amount of data is sufficiently large as to train an AI model and whether the data are sufficiently scalable to handle a large number of compounds.

Data Availability. It is important to consider whether the data are available in a digital format and whether it is possible to automate the data extraction process to facilitate integration with machine learning frameworks.

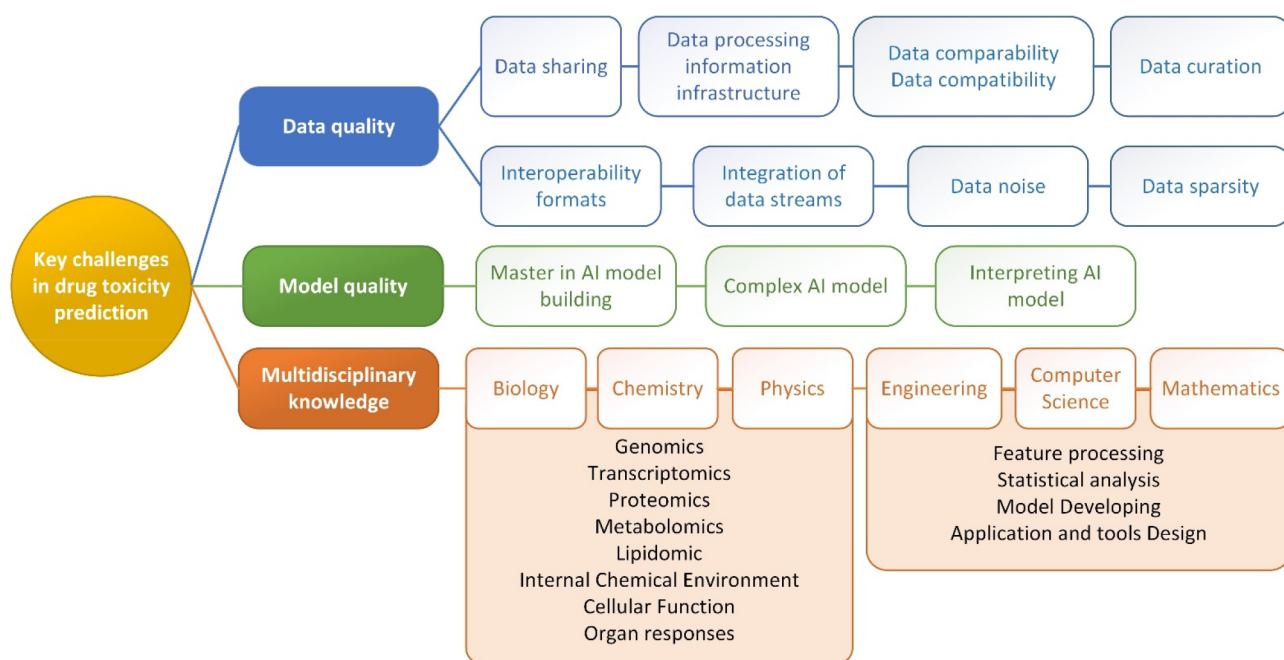


Figure 3. Key challenges in AI-based drug toxicity prediction.

Data Integration. It is important to consider whether the data, such as gene expression, proteomics, and metabolomics data, can be easily integrated with other databases or external data sources to enhance the predictive power of the model.

Data Preprocessing. It is important to consider whether the data are preprocessed and cleaned to remove noise, missing values, and errors, as these can affect the performance of the predictive models.¹⁶⁹

Furthermore, selecting the right database for drug toxicity prediction using AI is a crucial step, and time must be taken to evaluate different options, conduct pilot experiments, and evaluate different subsets of data to find the best fit for a specific use case. Supporting Information Table S2 presents a list of public databases and data sets widely used in toxicity prediction research. Additionally, even with the best-quality data, developing accurate predictive models for drug toxicity prediction is a challenging task, and other factors such as model complexity, feature selection, and validation strategies also play important roles.

4. DRUG TOXICITY PREDICTION TOOLS

In recent years, many advancements have been made in the ADMET and toxicity prediction tools. Supporting Information Table S3 presents a list of useful, free, and recently updated tools. These tools help predict the toxic properties of a large number of compounds and establish ADMET profiles and are important benchmarks for evaluating and comparing the scores of future models. Although the reported performance and prediction accuracy of most tools is good, performances for some properties remain low owing to a lack of data. The tools cannot be compared in terms of performance as their training and evaluation data sets cannot be standardized. However, researchers should select tools with high performance and large, reliable data sets. Moreover, researchers require numerous toxicity prediction tools to ensure that they make informed choices.

5. CHALLENGES AND FUTURE DIRECTIONS FOR AI-BASED DRUG TOXICITY PREDICTION RESEARCHERS

In silico drug toxicity prediction using AI has advanced significantly in recent years. However, certain challenges and limitations must be overcome. The main challenges in predicting drug toxicity are shown in Figure 3.

Insufficient high-quality data is one of the greatest challenges in using AI-based methods to predict drug toxicity.^{1,8–10} Developing a high-performance AI model requires both quantity and quality of training data. Although the sources of public toxicity data are increasing in quantity, the data quality remains limited. The current situation has evolved significantly from a small quantity of toxicology data hindering AI model development to data overload.¹⁷⁰ Although the volume of toxicological data is still less than that of big data in other disciplines, such as finance and economics, it is rapidly increasing and gradually forming big data. Collaboration for preparing big data presents many opportunities and challenges for toxicology prediction researchers.^{170,171} Handling large volumes of data poses many challenges in terms of data storage, retrieval, sharing, and processing, with data generation rates surpassing storage, analysis, and processing capabilities. However, big toxicology data sets with high complexity, including data diversity and variability, significantly improve the prediction performance. Therefore, toxicity data sets and repositories should cover a wide variety of data, including in vitro or chemical assay results, in silico predictions, genomic arrays, omics indices for various toxicity end points, epidemiology, surveillance, and exposure data.¹⁷² The creation of a big and diverse toxicity database encompassing a wide range of data resources necessitates the development of methods to standardize, manage, and integrate various sources of toxicological data which creates additional obstacles. Additional factors to consider when choosing a database to predict AI-based drug toxicity are discussed in Section 3.

Another challenge is the model quality. Several toxicity prediction models have performance results that are below 80%. Although improving model quality for toxicity prediction can be a challenging task, several approaches can be explored to improve the performance of these models. Expanding the scope and diversity of the training data can enhance the ability of a model to recognize a wider range of patterns, thereby improving its predictive capabilities. This strategy agrees with the preceding challenge in which relevant features or predictors are selected based on either domain expertise or statistical methods to steer the model toward key features that influence toxicity prediction. In addition, exploring sophisticated AI algorithms, such as transformers, may prove advantageous as they possess greater capacity for learning complex interrelationships between variables and excel at specific types of toxicity prediction tasks. Ensemble techniques can further improve overall model performance by minimizing both the variance and bias associated with individual models. Finally, seeking feedback from experts in the field would aid in identifying shortcomings and obtaining valuable guidance on strategies to refine existing models.

Many current AI models lack interpretability, making it difficult to understand how and why a model makes a particular prediction. This can be a barrier to improving the model performance and applying these models during drug development. There is a problem to solve the “black box” model to make it more interpretable.^{173,174} Therefore, the mechanistic interpretability and predictivity of the model should be balanced.¹⁷⁵ The utilization of model-agnostic techniques for interpretability, such as LIME,^{176–178} SHAP,¹⁷⁹ and Anchors¹⁸⁰ can provide an explanation for toxicity prediction. These methods allow for both global and local explanations while also identifying crucial features that are relied upon by the model to make predictions. Additionally, explainable neural networks (XNNs)^{181,182} should be considered because they incorporate explainable layers that highlight significant features through attention and gating mechanisms. The integration of these components can provide a rationale for supporting the predictive outcome of the model.

Another challenge faced by researchers is the limited mechanistic understanding of toxicity because toxicity involves multidisciplinary knowledge. Researchers in toxicity prediction need to have related knowledge in fields such as biology, chemistry, physics, engineering, computer sciences, and mathematics.¹⁰ Cross-disciplinary collaboration among researchers is a viable solution to overcome the challenge of limited mechanistic understanding of toxicity. Bringing together experts from different domains facilitates the exchange of knowledge and skill sets, identification of knowledge gaps, and formulation of innovative approaches to understanding toxicity. Another solution is to enhance interdisciplinary training for researchers to equip them with a comprehensive understanding of various fields and develop the necessary competencies and expertise for toxicity prediction. Additionally, novel advancements in artificial intelligence and machine learning can help address the issue of limited mechanistic understanding. These technologies can integrate data from various sources and fields, identify patterns and associations, and predict toxicity with high precision.

6. CONCLUDING REMARKS

Toxicity prediction studies for drug candidates have a significant impact on drug discovery and development and

can increase the speed and decrease the cost of drug creation which increases success rates in clinical trials. AI has the potential to improve the efficiency and accuracy of in silico drug toxicity prediction; however, addressing challenges and incorporating future perspectives are crucial for realizing this potential. Future studies should focus on developing more robust and interpretable AI models and integrating multiple data types and sources, such as imaging, genomics, and electronic health records. Explainable AI methods can also improve the transparency and reliability of AI-based predictions. Interdisciplinary collaborations among toxicologists, computer scientists, statisticians, and clinicians are likely to become increasingly important in the future, as they will bring together diverse perspectives and expertise to address the complex challenges of drug toxicity prediction. AI-assisted drug toxicity research is still in its early stages and further research is needed to assess its potential clinical significance.

In this review, we provide a comprehensive overview of the recent advances in AI-based drug toxicity prediction of six key toxicity properties, such as hERG, LD₅₀, DILI, Ames mutagenesis, carcinogenesis, skin sensitization, as well as Tox21 assay end points, and identify key areas for challenges and future research and development. These advances demonstrate the potential of AI in improving the drug development process and identifying potential toxicities earlier, which can reduce the need for animal testing and improve the safety of drugs for human use. However, comparison of the performance of AI models is difficult because of different tasks and data sets involved. No single model performed best in most cases. The performance of the dependent AI models was compared only when they were developed for the same toxicity end point and used the same data set for evaluation. Additionally, we provide a list of public data sources and toxicity prediction tools that can be useful to the research community. Hopefully, this review can serve as a useful reference for researchers interested in AI-based drug toxicity prediction models.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00200>.

Summary of recent advances in Tox21 assay end points prediction using AI, sources of public data for AI-based drug toxicity prediction, and public tools for AI-based drug toxicity prediction (PDF)

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Notes

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